MIL-C-25/2E
21 June 1968
SUPERSEDING
MIL-C-25/2D
29 September 1960

MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, PAPER-DIELECTRIC, DIRECT CURRENT (HERMETICALLY SEALED IN METAL CASES), STYLES CP25, CP26, CP27, CP28, AND CP29

The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-25.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DO NOT USE FOR NEW DESIGN
FOR REPLACEMENT PURPOSES ONLY

REQUIREMENTS:

Dimensions: See figure 1 and table II.

Case and bracket material: Nonmagnetic. (However, seal header may be constructed of magnetic material.)

Design and construction: Capacitors shall be of the design, construction, and physical dimensions specified on figure 1 and table I. In addition, brackets for styles CP27 and CP28 shall be approximately centered on the capacitor case and shall be riveted so as not to be easily shifted along the capacitor. Brackets for style CP29 shall be soldered or welded at the approximate center of the capacitor case and shall be wrapped around the capacitor a minimum of 180°

Capacitance value: See table I.

Capacitance tolerance: See table I.

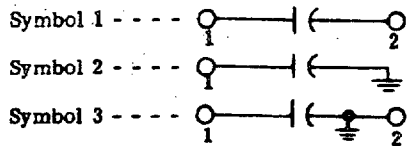
DC voltage rating: See table I.

Operating temperature range: -55° to +85° C.

Vibration grade: 1 (10 to 55 Hz, incl.)

Terminal: Symbol A - - - - - Axial wire lead.

Circuit diagram:



Voltage derating with temperature: See figure 2.

Dielectric withstanding voltage: Method 301 of MIL-STD-202.
Terminal to terminal - As specified in table III.
Terminal to case (when case is not a terminal) - As specified in table III.
There shall be no permanent damage, or open- or short-circuiting.

Insulation resistance: Method 302 of MIL-STD-202.
At 25° C:
Terminal to terminal - As specified in table IV.
Terminal to case (when case is not a terminal) - Shall exceed 3,000 megohms.
At 85° C:
Terminal to terminal - As specified in table IV.

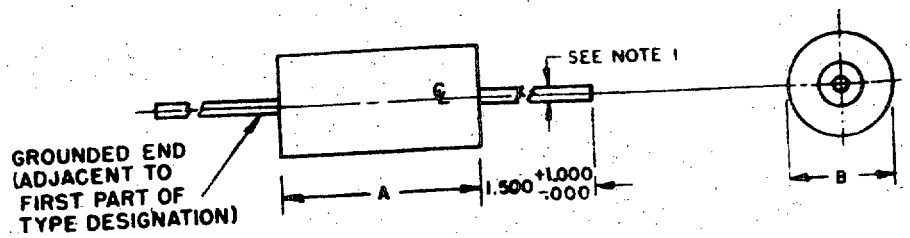
Capacitance: Method 305 of MIL-STD-202.
Measured capacitance shall be within the tolerance specified in table I.

Dissipation factor: When measured as specified in MIL-C-25, shall not exceed 1 percent.

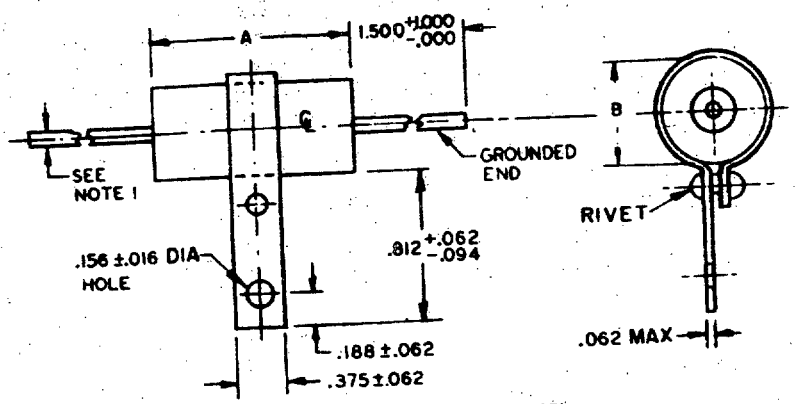
Barometric pressure (flashover): Method 105 of MIL-STD-202.
Test condition - B (50,000 feet).
Test potential - 125 percent of rated dc voltage applied for a minimum of 1 minute, without visible damage, external flashover, or open- or short-circuiting.

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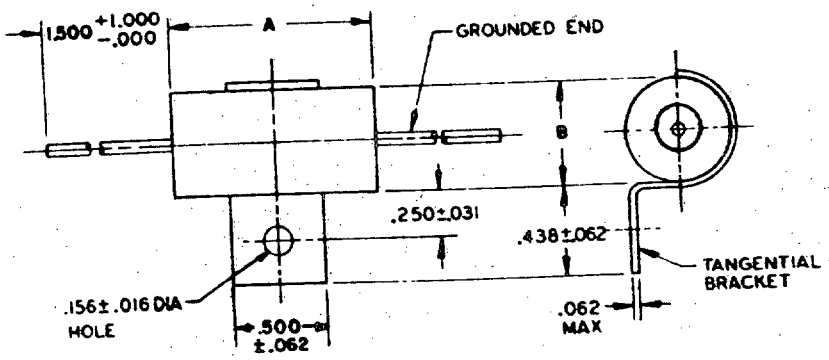


STYLE CP25 (UNINSULATED CASE)
STYLE CP26 (INSULATED CASE)



STYLE CP27 (UNINSULATED CASE)
STYLE CP28 (INSULATED CASE)

INCHES	MM
.016	.41
.031	.79
.040	1.02
.062	1.57
.094	2.39
.156	3.96
.188	4.78
.250	6.35
.375	9.53
.438	11.13
.500	12.70
.812	20.62
1.000	25.40
1.500	38.10



STYLE CP29 (UNINSULATED CASE)

NOTES:
1. Leads shall be of tinned solid wire .040 (No. 18 AWG).
2. Basic case dimensions for the insulated styles shall be the same as for the uninsulated styles.
3. See table I for additional dimensions.
4. All dimensions in inches.
5. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

FIGURE 1. Styles CP25, CP26, CP27, CP28, and CP29 capacitors.

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TABLE I. Styles CP25, CP26, CP27, CP28 and CP29 characteristics and dimensions.

Type designation 1/	DC voltage rating (see fig. 2)	Capacitance	Capacitance tolerance 2/	Characteristic	Vibration grade	Case dimensions - Styles CP25, CP27, CP29					
						Circuit 1		Circuit 2		Circuit 3	
						B±0.062	A +0.062 -0.125	B±0.062	A +0.062 -0.125	B±0.062	A +0.062 -0.125
	Volts	µf									
CP25A2EC104K3	200	0.1	K	E	1	---	---	0.688	1.500	---	
CP25A2EF104K3	600	0.1	K	E	1	---	---	0.688	1.750	---	
CP2-A-EH302K3	1,500	0.003	K	E	1	1.000	1.250	---	---	1.000	1.125
CP2-A-EH602K3	1,500	0.006	K	E	1	1.000	1.250	---	---	1.000	1.125
CP2-A-EH103K3	1,500	0.01	K	E	1	1.000	1.250	---	---	1.000	1.125
CP2-A-EH203K3	1,500	0.02	K	E	1	1.000	1.625	---	---	1.000	1.500
CP2-A-EH503K3	1,500	0.05	K	E	1	1.000	1.875	---	---	1.000	1.750
CP2-A-EH104K3	1,500	0.10	K	E	1	1.000	2.125	---	---	1.000	2.000

TABLE I. Styles CP25, CP26, CP27, CP28 and CP29 characteristics and dimensions - Continued.

Type designation 1/	DC voltage rating (see fig. 2)	Capacitance	Capacitance tolerance 2/	Characteristic	Vibration grade	Case dimensions - Styles CP26, CP28			
						Circuit 1		Circuit 3	
						B±0.094	A +0.188 -0.250	B±0.094	A +0.188 -0.250
	Volts	µf							
CP2-A-EH302K3	1,500	0.003	K	E	1	1.062	1.438	1.062	1.312
CP2-A-EH602K3	1,500	0.006	K	E	1	1.062	1.438	1.062	1.312
CP2-A-EH103K3	1,500	0.01	K	E	1	1.062	1.438	1.062	1.312
CP2-A-EH203K3	1,500	0.02	K	E	1	1.062	1.812	1.062	1.688
CP2-A-EH503K3	1,500	0.05	K	E	1	1.062	2.062	1.062	1.938
CP2-A-EH104K3	1,500	0.10	K	E	1	1.062	2.312	1.062	2.188

1/ Complete type designation will include additional symbols to complete style designation and to indicate circuit as applicable.

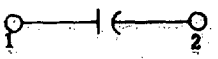
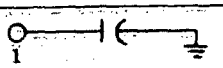
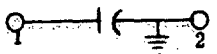
2/ Capacitance tolerance in percent: K = ± 10.

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TABLE II. Millimeter equivalents of decimal inches.

In	mm	In	mm	In	mm
0.062	1.157	1.125	28.58	1.812	46.02
0.094	2.39	1.250	31.75	1.875	47.63
0.125	3.18	1.312	33.32	1.938	49.23
0.188	4.78	1.438	36.53	2.000	50.80
0.250	6.35	1.500	38.10	2.062	52.37
0.688	17.48	1.625	41.28	2.125	53.98
1.000	25.40	1.688	42.88	2.188	55.58
1.062	26.97	1.750	44.45	2.312	58.72

TABLE III. Dielectric withstanding voltage test.

Circuit symbol	Circuit diagram	Test	Test connections	Test voltage (percent of rated dc voltage	Time test applied (minutes)
1		Terminal to terminal	1 to 2	200 1/	1 2/
		Terminal to case	1 and 2 to case	400 5/	1 3/ 4/
				200 + 1,000 6/ volts	1 3/ 4/
2		Terminal to terminal	1 to case	200 1/	1 2/
3		Terminal to terminal	1 to 2 or 1 to case	200 1/	1 2/

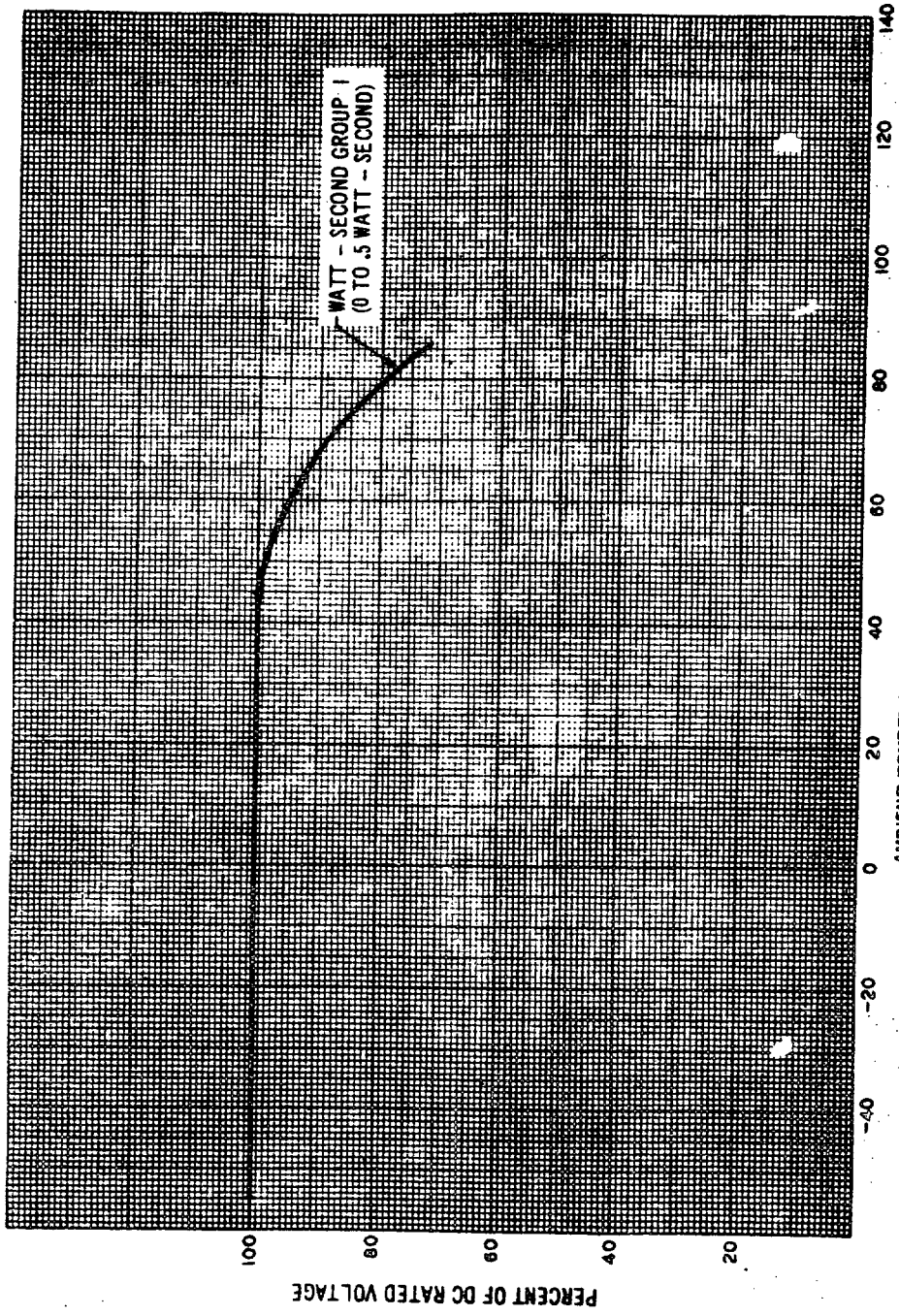
1/ 175 percent rated dc voltage after temperature-and immersion-cycling and moisture-resistance tests.
2/ For quality conformance inspection, and at the option of the supplier, either 200 percent of rated dc voltage shall be applied for 15 seconds, or 250 percent of rated dc voltage for not less than 1 second.
3/ For quality conformance inspection, the period of test-voltage application shall be 1 second.
4/ For quality conformance inspection, applications may be made between each terminal individually and the case, at the option of the supplier.
5/ For capacitors having dc voltage ratings of 600 volts and less.
6/ For capacitors having dc voltage ratings greater than 600 volts.

TABLE IV. Terminal to terminal insulation resistance measurements.

Capacitance values	Minimum insulation resistance 1/ Characteristic E
0.33 microfarads and less	At 25° C 6,000 megohms At 85° C 600 megohms
0.033 microfarads and less greater than 0.033 microfarads	20 megohm-microfarads 2/

1/ A potential equal to the dc rated voltage or 500 volts dc, whichever is less, shall be applied.
2/ Product obtained by multiplying the capacitance in microfarads by the insulation resistance in megohms.

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NOTE: Permissible operating voltages, while based on incomplete data, are the capacitor suppliers' best estimate to provide a life expectancy of 8,800 hours of continuous operation at higher ambient temperatures. Longer life can be expected by operation at voltages lower than indicated on the curve; for example, a life expectancy of 44,000 hours may be obtained by operation at 70 percent of the voltage determined by use of this curve. Also, a life longer than 8,800 hours may be expected at the voltage determined by use of this curve if the high ambient temperature prevails for only a portion of the whole operating time.

FIGURE 2. Voltage derating for ambient temperature (characteristic E).

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Vibration (grade 1, 10 to 50 Hz, incl):
Method 201 of MIL-STD-202.
Electrical load - 125 percent of rated dc voltage.
Test duration - 3 hours in each of 3 mutually perpendicular directions.
Measurements - An electrical signal of 1 ± 0.2 kHz, at a level of 1 ± 0.5 volts shall be applied during the last cycle in each direction. There shall be no momentary arcing or breakdowns, open- or short-circuiting, or mechanical damage.

Salt spray (corrosion): Method 101 of MIL-STD-202, test condition B (48 hours), with no evidence of corrosion, unwrapping or damage to sleeves, or obliteration of marking.

Temperature cycling: Method 102 of MIL-STD-202, test condition D.

Immersion: Method 104 of MIL-STD-202, test condition C, except that the duration of each immersion shall be 30 minutes, and change between solutions shall be made in not more than 3 seconds.
Measurements after final cycle:
Dielectric withstanding voltage at 25° C:
Insulating sleeves - Not less than 4,000 volts dc.
Terminal to terminal - As specified in table III.
Terminal to case (when case is not a terminal) - As specified in table III.
Insulation resistance at 25° C:
Insulating sleeves - Not less than 100 megohms.
Terminal to terminal - Not less than 30 percent of the value specified in table IV.
Terminal to case (when the case is not a terminal) - Not less than 1,500 megohms.

Solderability (axial wire leads only): Method 208 of MIL-STD-202.
Number of terminals to be tested - 2.

Terminal strength: Method 211 of MIL-STD-202, test conditions A and D.
Test condition A (pull): Applied force - 4-1/2 pounds, 1 pull.
Test condition D (twist): 3 rotations.

Moisture resistance: Method 106 of MIL-STD-202.
Polarization voltage - 100 volts.
Final measurements:
Dielectric withstanding voltage and insulation resistance requirements - As specified in immersion.

Seal: Method 112 of MIL-STD-202, test condition A, except that bath temperature shall be $+85\ ^{+3}_{-0}$ °C.
Liquid filled capacitors - 10 minutes at $+85°\ ^{+3}_{-0}$ °C on each of two opposite sides with no evidence of leakage.

Low temperature and capacitance change with temperature:
Low temperature - .
Conditions: -55° ± 3°C for 58 ± 4 hours with rated dc voltage applied.
Capacitance change with temperature - Not to exceed ± 10 percent.

Life: Method 108 of MIL-STD-202, test condition B ($250\ ^{+8}_{-0}$ hours).
Electrical load:
Group IA (0.08 watt-second and less), excepting 1,000 volt capacitors - 140 percent of dc rated voltage.
Group IB (greater than 0.08 watt-second), and up to and including 0.5 watt-second) and 1,000 volt capacitors - 120 percent of dc rated voltage.
Final measurements (at 25° C):
Insulation resistance:
Terminal to terminal - Not less than 30 percent of the value specified in table IV.
Capacitance: Within 10 percent of the initial value.
Dissipation factor: Change not more than 1 percent from initial value.

Inspection lot: An inspection lot shall consist of capacitors of the same characteristic covered by this specification sheet.

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Custodians:
Army - EL
Navy - EC
Air Force - 11

Review activities:
Army - EL, MU, MI
Navy - EC, AS, OS
Air Force - 11, 17, 85
DSA - ES

User activities:
Army -
Navy -
Air Force - 19

Preparing activity:
Air Force - 11

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